Quantitative Data Analysis

Session 4 — What Drives My Outcome? Simple & Multiple Regression

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EM Normandie — Programme Grande École
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Where we are

Session	Method	Question it answers
1	Univariate descriptive analysis	What does my data look like?
2	Chi-square, correlation & t-tests	Are two variables related? Do two groups differ?
3	ANOVA + post-hoc tests	Do groups differ from each other?
4	**Simple & multiple regression**	**What drives my outcome?**
today		

> Session 2 told us visits and spending are correlated ($r = 0.62$). Session 3 told us store format matters ($\eta^2 = 0.11$).
> Today we put everything in one model — and put a **euro value** on each driver.

Today: 3 hours 30

Time	What happens	Mode
0:00 – 0:15	Recap of Session 3	Plenary
0:15 – 0:50	**Simple regression:** from a correlation to a line	Lecture
0:50 – 1:30	**Multiple regression:** several drivers, all else equal	Lecture + quiz
1:30 – 1:45	*Break*	
1:45 – 2:10	**Conditions and diagnostics:** can I trust the model?	Lecture
2:10 – 3:00	**Jamovi lab:** your regression model	Teams of 3
3:00 – 3:30	From four analyses to one recommendation: the final report	Workshop

The last session of the fil rouge. You leave with a **model of your outcome** and the plan of the consulting report that closes the course.

Simple regression: from a correlation to a line

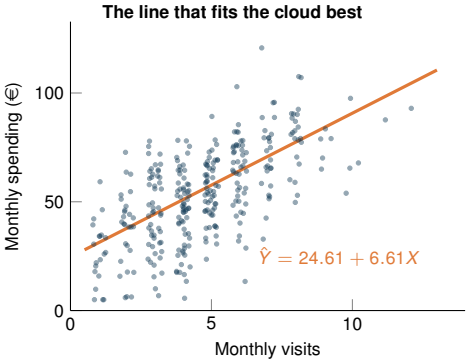
The question

Session 2: visits and spending are correlated, $r = 0.62$.

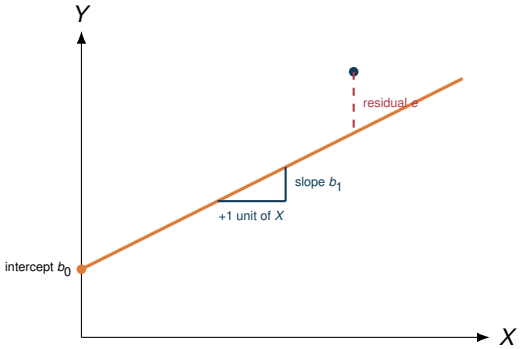
The manager's next question: **how many euros is one more visit worth?**

Correlation is **symmetric** and has **no unit**.

Regression is **directional** ($X \to Y$) and speaks in the **units of Y**: euros, tonnes, percentage points.



Anatomy of a regression line



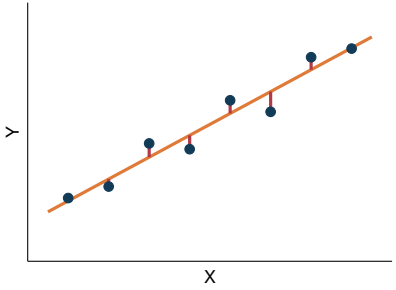
$$Y = b_0 + b_1 X + e$$

b_0 **(intercept):** predicted Y when $X = 0$. Often meaningless on its own.

b_1 **(slope):** change in Y for one more unit of X. **The number managers want.**

e **(residual):** observed minus predicted. What the line does not explain.

Least squares: how the line is chosen



Each **red segment** is a residual: the prediction error for one observation.

The regression line is the one that makes the **sum of squared residuals as small as possible** — hence “ordinary least squares” (OLS).

> You will never compute it. You need to know that a residual is **the part of each customer’s spending the model gets wrong** — we will inspect them later.

Worked by hand: 8 stores NEW

Advertising x (k€) vs sales y (k€), 8 stores (Session 2):

	Advertising x	Sales y
Mean	5.50	38.875
SD	2.449	6.289
Correlation r	0.969	

$$b_1 = r \times \frac{s_y}{s_x} = 0.969 \times \frac{6.289}{2.449} = 2.49$$

$$b_0 = \bar{y} - b_1 \bar{x} = 38.875 - 2.49 \times 5.5 = 25.19$$

Sales $= 25.19 + 2.49 \times$ **advertising**

Each extra 1 k€ of advertising is associated with **2.49 k€** more sales.

$R^2 = r^2 = 0.94$.

The slope is the correlation **translated into units**: r says "how tightly", b_1 says "how much". Same sign, always.

The Jamovi output

Model Fit Measures

R	R^2	Adj. R^2	F	df1	df2	p
0.618	0.383	0.381	200.1	1	323	$< .001$

Model Coefficients — Spending

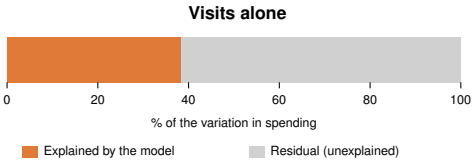
Predictor	Estimate	SE	t	p
Intercept	24.61	2.27	10.85	$< .001$
Visits	6.61	0.47	14.14	$< .001$

H_0: visits have no effect on spending ($b_1 = 0$).
$t = 14.14$, $p < .001 \rightarrow$ reject H_0.

> Each additional monthly visit is associated with **6.61 € more spending** per month.

In simple regression, $R = |r|$ and $R^2 = r^2$: the 38% of Session 2 again.
Check: $t = B/SE = 6.61/0.47 = 14.1$ ✓

R^2: how much of the outcome the model explains

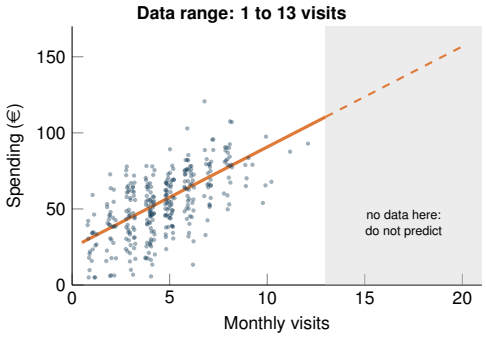


R^2 ranges from 0 (the model explains nothing) to 1 (it explains everything).

R^2	Reading (management data)
< 0.10	weak
$0.10 – 0.40$	moderate
$0.40 – 0.70$	substantial
> 0.70	strong

62% of spending is unexplained by visits. Something else matters — income, the loyalty card, distance? One predictor is rarely enough.

Predicting — and the danger of extrapolation



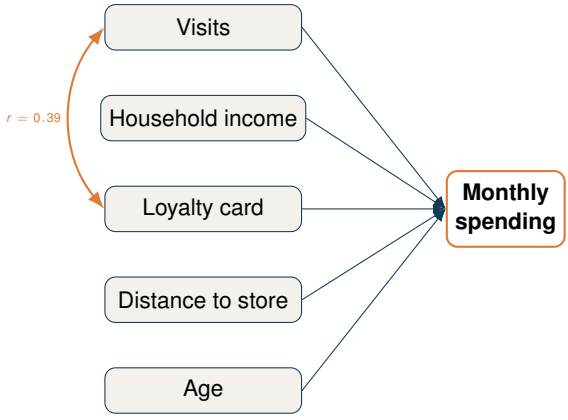
Prediction for a customer with 5 visits:

$$\hat{Y} = 24.61 + 6.61 \times 5 = 57.66\,€$$

> "A customer with 20 visits would spend 157 €" — **no**. No one in the data visits more than 13 times. Outside the observed range, the line is a guess.

Multiple regression: all else being equal

Why several predictors?



Spending has several drivers, and the drivers are **related to each other**: card holders also visit more often ($r = 0.39$).

In a simple regression, the “visits” effect silently includes part of the card effect.

> Multiple regression estimates the effect of each driver **holding the others constant** — *ceteris paribus*.

The model

$$Y = b_0 + b_1 X_1 + b_2 X_2 + \cdots + b_k X_k + e$$

Y — the outcome
Quantitative (interval / ratio): spending, profit, emissions.

b_0 — intercept
Predicted Y when every $X = 0$. Rarely a real situation.

b_j — coefficients
Change in Y for $+1$ unit of X_j, **the other X held constant**.

e — error
Everything the model does not capture. It exists in every estimation.

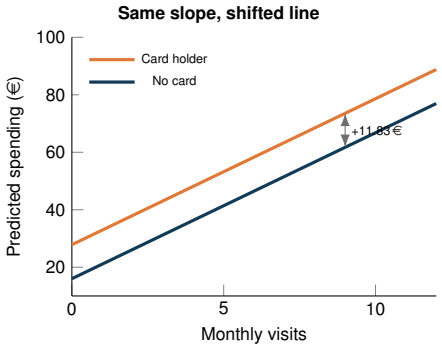
Predictors can be quantitative or qualitative — if you code them as dummy variables.

Qualitative predictors: dummy variables

Loyalty card: 1 = holder, 0 = no card.
Its coefficient = the **gap between the two groups**, all else equal.

A variable with k categories needs $k - 1$ **dummies**. The missing one is the **reference**: store format $\rightarrow$ Super, Conv., Online vs Hyper (reference).

> Jamovi codes this for you: put the variable in **Factors**, choose the reference level.



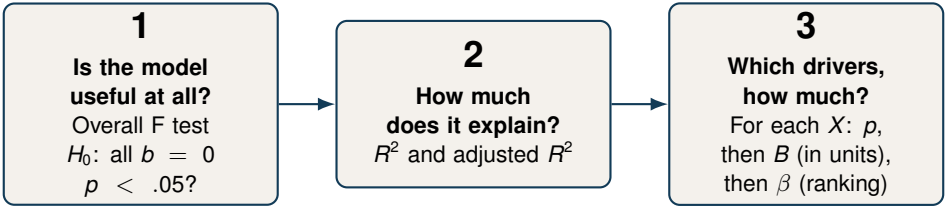
The case: our model of monthly spending

	Variable	Type	Unit
Y	Monthly spending	Continuous	€ per month
X_1	Monthly visits	Discrete	visits
X_2	Household income	Continuous	k€ per year
X_3	Loyalty card	Dummy (0/1)	1 = holder (39%)
X_4	Distance to store	Continuous	km
X_5	Age	Continuous	years

Managerial question: which levers raise a customer's monthly spending, and by how much?

Sample size rule: at least 10–15 observations per predictor. $N = 325$ for 5 predictors: comfortable.

Reading the output: three steps, in this order



If step 1 fails ($p \geq .05$), **stop**: the model as a whole explains nothing, and individual coefficients are not worth interpreting.

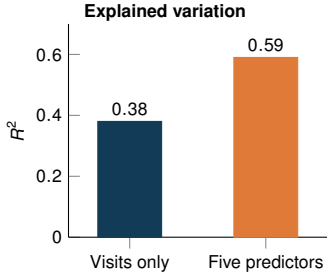
Steps 1 and 2: the model as a whole

Model Fit Measures

R	R^2	Adj. R^2	F	df1	df2	p
0.769	0.592	0.586	92.55	5	319	$< .001$

Step 1. $F(5, 319) = 92.55$, $p < .001$: the model is significant. ($df1$ = number of predictors = 5; $df2 = N - k - 1 = 325 - 5 - 1 = 319$.)

Step 2. Together, the five predictors explain **59%** of the variation in monthly spending — up from 38% with visits alone.



Adjusted R^2 penalises useless predictors: report it when comparing models.

Step 3: the coefficients

Model Coefficients — Monthly spending (Jamovi)

Predictor	Estimate (B)	SE	t	p	Stand. (β)	VIF
Intercept	−8.50	6.14	−1.38	0.167		
Visits	5.08	0.42	12.10	$< .001$	0.475	1.20
Income (k€)	0.98	0.10	9.36	$< .001$	0.345	1.06
Loyalty card: Yes – No	11.83	2.02	5.86	$< .001$	0.233	1.23
Distance (km)	−0.75	0.30	−2.47	0.014	−0.090	1.04
Age	0.03	0.10	0.31	0.759	0.011	1.01

Four significant drivers ($p < .05$): visits, income, loyalty card, distance.

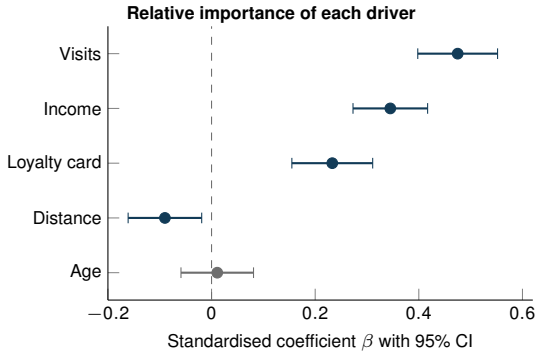
Age is not significant ($p = .759$) — consistent with Session 2 ($r = -0.05$).

Interpreting B: always in the units, always "all else equal"

Driver	Interpretation
Visits ($B = 5.08$)	One more visit per month $\rightarrow$ **+5.08 €** of monthly spending, for customers with the same income, card status, distance and age.
Income ($B = 0.98$)	+1 k€ of annual household income $\rightarrow$ **+0.98 €** per month.
Loyalty card ($B = 11.83$)	Card holders spend **11.83 € more** per month than comparable non-holders.
Distance ($B = -0.75$)	Each extra km between home and store $\rightarrow$ **$-$0.75 €** per month.
Intercept (-8.50)	Customer with 0 visits, 0 income, 0 km, aged 0: meaningless. **Do not interpret.**

Why did visits drop from 6.61 € to 5.08 €? In the simple model, visits also carried part of the loyalty-card effect. Controlling for it gives the **own effect** of a visit.

Which driver matters most? Standardised coefficients



B values have different units (visits, k€, km): **you cannot compare them**.

β puts every variable in **standard deviations**: now they are comparable.

> Ranking: visits (0.48) $>$ income (0.35) $>$ card (0.23) $>$ distance (-0.09). A CI crossing 0 (age) = not significant.

In a sensible model $|\beta|$ is almost always below 1.

Quiz — interpret it right

1. A colleague writes: “Card holders spend 11.83 € more, so the card *causes* 11.83 € of spending.” What is the problem?
2. “Visits have the biggest effect because 5.08 is bigger than 0.98.” True or false?
3. “Age has no effect on spending.” How should it be phrased?
4. Can we predict the spending of a customer with 30 visits a month?
5. Adding 10 random variables to the model raises R^2 from 0.592 to 0.604. Good news?

Quiz — answers

#	Answer
1	Association, not causation: customers who spend a lot may choose to take a card. Say “is associated with”.
2	False reasoning — the units differ (visits vs k€). The right conclusion comes from β: 0.475 vs 0.345.
3	“We find no significant effect of age, once the other variables are controlled for ($p = .759$).”
4	No: the data go up to 13 visits. That is extrapolation.
5	No: R^2 always rises when you add variables. Check adjusted R^2 — it will fall.

Conditions: can I trust the model?

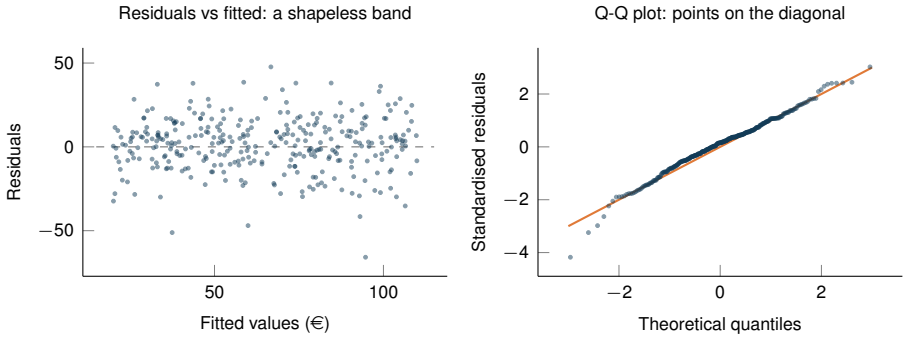
The conditions of linear regression

1. **Linearity** — each X relates to Y roughly along a straight line (scatterplots).
2. **Independence of residuals** — Durbin–Watson close to 2 (between 1.5 and 2.5).
3. **Normality of residuals** — Q-Q plot of residuals along the diagonal.
4. **Equal variance of residuals** (homoscedasticity) — no funnel in the residual plot.
5. **No multicollinearity** — predictors not too correlated with each other: VIF < 5.

Note: the condition is on the **residuals**, not on Y itself. A skewed Y can still give well-behaved residuals.

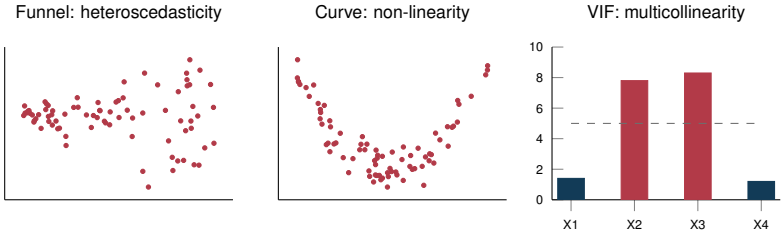
Our case: Durbin–Watson = 1.83; all VIF ≤ 1.23; residuals checked on the next slide.

Diagnostics: the two plots to look at



No pattern, no funnel, points along the line: **the conditions hold**. (Shapiro–Wilk on the residuals: $p = .349$.)

What a problem looks like



Problem	What to do
Funnel	Log-transform Y (turnover, emissions); report robust conclusions with caution
Curve	Log-transform X, or add X^2
VIF > 5	Drop one of the correlated predictors (e.g. keep turnover or headcount, not both)

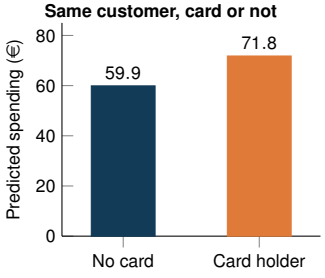
Prediction with the full model

Customer profile: 6 visits, income 40 k€, card holder, lives 3 km away, aged 41.

$$\hat{Y} = -8.50 + 5.08(6) + 0.98(40) + 11.83(1)$$

$$-0.75(3) + 0.03(41) \approx 71.8\,€\text{per month}$$

(Computed with unrounded coefficients.)



This is the language of a business case: "**converting this customer to the card is worth about 12 € a month, 142 € a year**."

Writing it up — model answer

1. IDENTIFY. Monthly spending is quantitative and we want the simultaneous effect of several predictors: multiple linear regression. Residual plots show no pattern, the Q-Q plot is on the diagonal, Durbin–Watson = 1.83, all VIF ≤ 1.23: the conditions hold.

2. HYPOTHESISE. Overall: H_0, none of the predictors explains spending. For each predictor: H_0, it has no effect once the others are controlled for.

3. DECIDE. The model is significant, $F(5, 319) = 92.55$, $p < .001$, and explains 59% of the variation (adjusted $R^2 = 0.586$). Visits ($B = 5.08$, $p < .001$), income ($B = 0.98$, $p < .001$) and the loyalty card ($B = 11.83$, $p < .001$) increase spending; distance reduces it ($B = -0.75$, $p = .014$). Age has no significant effect. By standardised coefficients, visits are the strongest driver ($\beta = 0.48$).

4. RECOMMEND. (1) Push visit frequency — each extra monthly visit is worth about 5 €: weekly offers, app notifications. (2) Grow the loyalty card base: +12 € per holder per month, all else equal; test whether a sign-up bonus below that value pays off. (3) Stop age-based targeting: it adds nothing once behaviour is known.

Read output critically — a real example

Regression of monthly sales (k€) on price and advertising, 48 stores

R	R^2	Adj. R^2
0.79	0.62	0.71

Predictor	Estimate	SE	t	p
Intercept	52.10	8.40	6.20	$< .001$
Price (€)	-4.20	2.10	-5.60	$< .001$
Advertising (k€)	1.30	0.40	3.25	0.002

Two things are wrong here.
Can you spot them before I tell you?

Adj. R^2 = 0.71 $>$ R^2 = 0.62 — impossible: the adjustment can only lower R^2.
$t = -5.60$ **for price** — $t = B/SE = -4.20/2.10 = -2.00$. The p-value built on it is wrong too.

Checks for every regression: Adj. $R^2 \leq R^2$, $t = B/SE$, $R^2 = R \times R$, and signs that make business sense.

Jamovi lab

Linear regression in Jamovi

Analyses → Regression → Linear Regression

1. **Dependent Variable:** your quantitative outcome
2. **Covariates:** continuous predictors; **Factors:** qualitative ones (set the Reference Levels)
3. **Model Fit:** R, R^2, Adjusted R^2, F test
4. **Model Coefficients:** Standardized estimate, confidence interval
5. **Assumption Checks:** Autocorrelation test, Collinearity statistics, Q-Q plot of residuals, Residual plots
6. Optional: **Save** → predicted values, residuals

Read, in this order:

1. Assumption checks — trustworthy?
2. F test — useful at all?
3. R^2 / adjusted R^2 — how much?
4. Each p, then B in units
5. β — ranking of drivers

Firm data: log-transform skewed variables first (Data → Compute: `LN(Turnover)`). Then B reads approximately as a % effect.

Apply it to your fil rouge dataset

Firm-level data
Y: net profit, or ROI
X: turnover (log), headcount, debt ratio, emissions per employee, sector (factor)
Do greener firms perform better, all else equal?

Customer data
Y: spending, or satisfaction score
X: visits, income, distance, segment (factor)

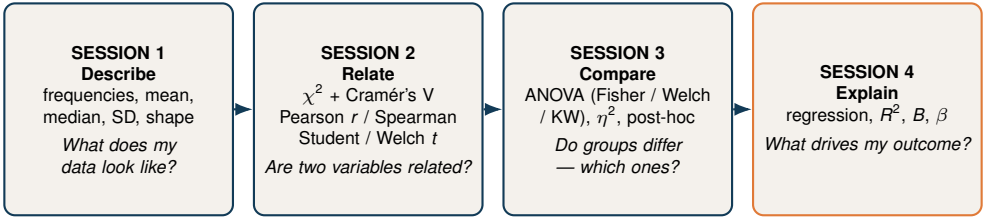
Build the model in three moves:

1. Start from the predictors your Session 2 correlations flagged
2. Add the grouping variable of Session 3 as a factor
3. Check VIF: two predictors correlated at $r > 0.8$? Keep one.

Aim for a model a manager can read: 3 to 6 predictors, each with a reason to be there.

From four analyses to one recommendation

The whole course on one slide



One template for all of it: **IDENTIFY** $\rightarrow$ **HYPOTHESISE** $\rightarrow$ **DECIDE** (significance and strength) $\rightarrow$ **RECOMMEND**.

Revision table: one line per method

Method	Variables	H_0	Report	Strength
Chi-square	2 qualitative	Independent	$\chi^2(df), p$	Cramér's V
Pearson / Spearman	2 quantitative (or ordinal)	No correlation	$r(df), p$	$\lvert r\rvert$, r^2
Student / Welch t	Quantitative Y, 2 groups (Levene decides)	Equal means	$t(df), p$	Cohen's d
One-way ANOVA	Quantitative Y, 3+ groups	All means equal	$F(df_1, df_2), p$ + post-hoc	η^2
Fisher + Tukey	Levene $p > .05$			
Welch + Games–Howell	Levene $p \leq .05$			
Kruskal–Wallis + DSCF	small n, not normal			ε^2
Bonferroni	few planned pairs		$p \times m$	
Multiple regression	Quantitative Y, several X	All $b = 0$ / each $b = 0$	F, p; each B, p	R^2, adj. R^2, β

Opposite logic ($p > .05$ = good news): Shapiro–Wilk (normality) and Levene (equal variances).

Always **three numbers**: the statistic with its df, the p-value, and the effect size.

The final consulting report

1. **Executive summary** — 1 page, 3 recommendations
2. **Data** — source, provenance, cleaning
3. **Profile** — univariate (S1)
4. **Relationships** — χ^2, r, t-test (S2)
5. **Group differences** — ANOVA (S3)
6. **Drivers** — regression (S4)
7. **Limits** — what the data cannot say

Write the executive summary last — but put it first.

Each recommendation must be:

- **traced** to a specific result ("$B = 11.83$, $p < .001$")
- **quantified** in business units (€, %, customers)
- **actionable** on Monday morning

Where two sessions disagree (e.g. a raw correlation vs a controlled regression coefficient), say so and **trust the model**. That is analysis.

Errors that cost marks every year

- Interpreting coefficients before checking the F test and the conditions
- Comparing raw B values across variables with different units — use β
- Forgetting “all else being equal” in every interpretation
- Interpreting a non-significant coefficient as a real effect
- Reading the intercept as a meaningful quantity
- Writing “causes” — observational data show associations
- Predicting outside the range of the data
- Putting a qualitative variable in Covariates instead of Factors

What you can now do

1. **Identify and justify** the appropriate statistical method for a managerial problem
2. **Implement** it in Jamovi
3. **Interpret** the output accurately — and spot when it is wrong
4. **Draw conclusions** and formulate actionable recommendations

These were the four objectives of Session 1.

You have now done each of them four times, on your own data.

> The software computed. **You** decided, interpreted, and recommended.

Questions?

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Final report due as announced on the course page.

Good luck!